

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):		
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)	
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)	
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)	
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)		
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.		
ê by polarization plug into templates above		
Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{ } = (1 + \Gamma_{ }) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{ } ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_2}{\eta_1}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{ } ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{ } = 1 - R_{ }$
Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; <i>nonmag.</i> $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.			
Intrinsic impedance η_i plug into table above			
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$			
Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.			
Low-loss correction $\eta_c \propto \sigma / (\omega \epsilon) \ll 1$			
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$			
For Γ/τ just use $\eta_0 / \sqrt{\epsilon_r}$. The j term is for $ \eta_c , \theta_{\eta}$. See LOSSY MEDIA — 5 CASES.			

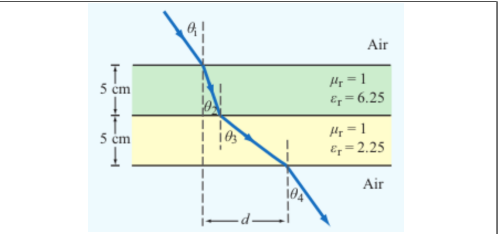
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	
$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$	
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	
$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$	
Air-slabs-air shortcut <i>outer media identical</i>	
$\theta_{\text{exit}} = \theta_{\text{incident}}$	
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	$\eta = \eta_0 / \sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$
Low-loss	< 10 ⁻²	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon'}{2\omega \epsilon_r}\right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: $(\eta / \sqrt{X}) e^{j\theta_{\eta}}$ see below
Good cond.	> 10 ²	$\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1
 $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$
 $\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$
 $\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$ TM mode $H_z = 0, E_z \neq 0$
Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.
E_x form (TE) read m, n from arguments
 $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$
coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m=0 or n=0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode <i>generator: $\tilde{H}_z$</i>	
$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TE}, \tilde{H}_y = \tilde{E}_x / Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	
TM mode <i>generator: $\tilde{E}_z$</i>	
$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TM}, \tilde{H}_y = \tilde{E}_x / Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	
TEM plane wave <i>for reference</i>	
$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$	
$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A}, k = \omega \sqrt{\mu\epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k \sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3×10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
 $t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower u_g ⇒ arrive later.

BOUNDARY

η_i, θ_t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/TM — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
η = η₀/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√1 - (f_c/f)²
u_g = u_{p0}√1 - (f_c/f)²
u_pu_g = u_{p0}²
t = L/u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r, %P_t — pwr refl./trans. (unitless)
%P_r = 100|Γ|²
%P_t = 100|τ|² η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 — always start here λ = wavelength (m), l = antenna rod length (m), $c=3\times10^8$ m/s

$\lambda = \frac{c}{f}$ (m)	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
= 1	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin\theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2}{32\pi^2 R^2} \sin^2\theta = S_{\text{max}} \sin^2\theta$ (W/m ²)
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377\ \Omega$; $S_{\text{max}} = \text{max pwr density (the prefactor)}$.
Max @ broadside $\perp$ to dipole axis = equator plane $\theta_{\text{max}} = \pi/2$; D exact $D=1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

°	rad	sin θ	cos θ	sin ² θ	cos ² θ
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

N identical elements along axis, spacing d , equal phase (broadside, $\theta_{\text{max}} = \pi/2$). Total pattern $S = S_e \cdot F_a$ (S_e = single elem).
Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos\theta$
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
Normalize $F_a^{\text{norm}} = F_a/N^2$
HPBW (broadside, uniform) use Nd, not $(N-1)d$
$\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$
Grating lobes appear if $d > \lambda$. Avoid.

BEAM PARAMETERS — β , Ω_p , D

Normalize *always start here*

$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)

Beamwidth β at threshold X

Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p recognize pattern, look up
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin\theta \, d\theta \, d\phi$ (sr)
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin\theta \, d\theta$.
θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi \approx 12.57$
Hertzian sin ² θ	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_1^2/2 / \ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π .

Directivity D	<i>always $D = 4\pi/\Omega_p$; common shortcuts below</i>
$D = \frac{4\pi}{\Omega_p}$	(unitless) $D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction” = θ_{max} (rad); “directivity” = D (unitless).	
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$.	
Gain (if asked)	$\xi = \text{rad. eff.}, 0 < \xi \leq 1$
$G = \xi D$	$G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.	

COMMON DIPOLE PARAMETERS

		Lossless ($\xi=1$): $G=D$.
Type	l	D D_{dB} R_{rad} (Ω)
Isotropic	N/A	1 0 —
Hertzian	$< \lambda/50$	1.5 1.76 $80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64 2.15 73.2
Monopole	$\lambda/4$ (gnd)	1.64 2.15 36.6
Full-wave	λ	2.41 3.82 199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

ANTENNA AREAS (A_e , A_p)

Effective area antenna's RX cross section; D from BEAM PARAMS or table above
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section wire dipole, diameter d; l see table above
$A_p = l d$ (m ² , any dipole)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRIIS TRANSMISSION FORMULA

For each antenna's G : if given by NAME (“half-wave dipole” etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.

- Get G_t, G_r by antenna type:
 - named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$ lossless)
 - aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G = D \approx 4\pi A_p / \lambda^2$)
 - arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G = \xi \cdot 4\pi / \Omega_p$)
 - given in dB $\rightarrow$ dB $\leftrightarrow$ LINEAR table
- Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G , D , P_r/P_t , SNR , ...):

$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$		$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$	
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $\text{SNR} > 10$ dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); A_e effective area (m², $= \lambda^2 D / 4\pi$); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ , d aperture side / diameter (m).

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi / \beta^2$$

HPBW and D by aperture shape uniform illum.; β in rad			
Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi / (\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi / \beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi / \beta^2 = \pi^2 d^2 / \lambda^2$
Scaling rules $D \propto A_p / \lambda^2$; $\beta \propto \lambda / \sqrt{A_p}$			
Change	New D	New β	
Double ($A_p \rightarrow 2A_p$)	area $D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}} / \sqrt{2}$	
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$	

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $D = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\max} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a / N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)